Computer Science Department

Master Degree in Computer Science

PROJECT DOCUMENTATION for BIG DATA

Link Prediction in Social Networks

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1 Business Understanding

This chapter covers the business understanding phase of the CRISP-DM methodology.

1.1 Determine Business Objectives

Background

Social network analysis (SNA) has emerged as a critical pursuit in understanding the complex dynamics of modern online communities. While traditional statistical techniques remain valuable, SNA introduces specialized measures to capture the dependencies between social entities, characterize their behaviors, and assess their impact on the network both structurally and temporally [1]. The fundamental principle of SNA lies in shifting the analytical focus from individual social actors to the relationships and interactions that bind them, revealing patterns that emerge from the network's underlying structure rather than from isolated entities [1].

In the globalized digital landscape, social networks have become one of the favorite means for modern society to perform social interactions and exchange information via the Internet, allowing people to share their contents without boundaries [2, 3]. Through these platforms, individuals communicate and express their comments, likes, and interests in a fast and easy manner, creating complex connections that can be immediately visualized as large graphs [2]. This massive spread of social media has introduced unprecedented opportunities for information dissemination [4]. However, this very interconnectedness presents significant challenges: the same platforms that facilitate legitimate social connections also provide powerful tools for malicious activities, including misinformation campaigns, cyberbullying, hate speech propagation, and the radicalization of vulnerable individuals [5, 3, 4, 6].

Risky users exploit the spreading power of social networks for illegal or harmful activities such as inciting religious or political extremism, disseminating disinformation, or recruiting. For these reasons, detecting them has become a fundamental task for ensuring the safe use of these platforms [3, 5, 6]. This challenge is particularly acute when dealing with borderline users who appear risky from certain perspectives but not from others, and is further complicated by the dynamic nature of social networks [3, 5]. User relationships, content semantics, and behavioral patterns continuously evolve; models that treat networks as static entities fail to capture users who transition from safe to radicalized behavior over time [5].

The unstructured nature of user interactions on social media platforms renders manual identification of malicious actors both tedious and impractical at scale [4]. Recent research has increasingly focused on leveraging graph-based representations, where users constitute nodes

and their interactions form edges, to address this complexity [4, 7]. Link prediction, defined as the process of predicting future connections between pairs of unconnected links based on existing connections [2], represents a fundamental problem in network analysis that has evolved from static snapshot-based approaches to temporal methodologies that exploit multiple network states across time to predict future or re-occurring connections [1, 2]. This temporal perspective aligns with the recognition that real-world networks are not static but dynamic and continuously evolving, requiring analytical frameworks that can capture growth patterns and temporal dependencies [1].

Given the urgency of protecting society from exposure to discriminatory, hateful, and violent content at the massive scale of contemporary social networks, there is a critical need for effective automated systems to assist Law Enforcement Agencies in identifying and addressing users who disseminate malicious content [6]. This work addresses this need by developing methods that integrate multiple perspectives of user behavior, account for the temporal evolution of networks, and leverage the relational structure inherent in social media platforms to enable accurate and timely identification of risky users.

Business Goals

The primary business goal of this project is to develop an effective temporal link prediction system for social networks that can:

- Predict future connections between users based on their historical interactions and content
- Evaluate the effectiveness of incremental learning approaches for temporal network evolution
- Given a dataset of users and relative posts, generate realistic synthetic social network structures
- Provide insights into how network properties evolve over time

Business Success Criteria

The project will be considered successful if it achieves the following outcomes:

- Development of a temporal link prediction model based on EvolveGCN
- Quantitative comparison of model performance using standard metrics (Mean Average Precision, AUC, Precision, Recall, F1)
- Successful transfer learning application: trained models can generate plausible network structures on synthetic datasets
- Analysis of network evolution patterns and structural properties across temporal snapshots
- Generation of a large synthetic network of 1000 artificial users with realistic connection patterns

1.2 Assess Situation

Inventory of Resources

The available data are being collected from the social network platform Gab[1], known for its minimal content moderation policies and popularity among far-right communities. The dataset includes user profiles, posts, and interaction data (likes, comments, shares) over a defined time period.

Current resources available for the project are the following machines:

1. CPU: AMD Ryzen 7 7700, 8 cores, 16 threads; GPU: NVIDIA RTX 5060Ti, 16 GB VRAM; RAM: 48 GB
2. CPU: AMD Ryzen 7 7840HS, 8 cores, 16 threads; GPU: Integrated AMD Radeon 780M; RAM: 48 GB
3. CPU: Intel Core i7-10750H, 6 cores, 12 threads; GPU: NVIDIA GTX 1650 Mobile Max-Q, 4 GB VRAM; RAM: 16 GB

Identify Constraints

There are no significant constraints anticipated for this project. However, potential challenges include:

- Data Quality: The collected data may contain noise, missing values, or inconsistencies that could affect model performance.
- Computational Resources: Training complex models on large datasets may require significant computational power and time.
- Ethical Considerations: Handling user data requires adherence to privacy regulations and ethical guidelines to ensure responsible use of information.

Assumptions

- The dataset from Gab is representative of broader social network dynamics and user behaviors.
- A connection between users exists if they are connected in one snapshot
- Graphs are undirected and unweighted
- If a connection exists between two users in a snapshot, it will also exist in all subsequent snapshots

1.3 Determine Data Mining Goals

Data Mining Goals

Here we translate the business goals into specific data mining objectives:

[1] https://gab.com/

- Develop a temporal link prediction model using EvolveGCN to forecast future user connections based on historical interaction data.
- Evaluate the model's performance using metrics such as Mean Average Precision (MAP), Area Under the Curve (AUC), Precision, Recall, and F1-score.
- Adapt and test the trained model for generating synthetic social network structures.
- Analyze the temporal evolution of network properties, including degree distribution, clustering coefficient, and average path length.

Success Criteria

The success of the data mining efforts will be measured by:

- Achieving high performance on link prediction tasks
- Demonstrating the model's ability to generalize to synthetic datasets, producing realistic network structures.
- Providing comprehensive insights into the temporal dynamics of the social network through detailed analysis of evolving network metrics.

2 📖 Data Understanding

This chapter covers the data understanding phase of the CRISP-DM methodology.

2.1 Collect Initial Data

Collect initial data from relevant sources.

2.2 Describe Data

Describe the data structure, format, and characteristics.

2.3 Verify Data Quality

Verify data quality by checking for errors, missing values, and inconsistencies.

2.4 Explore Data

Explore data to gain initial insights and identify patterns.

3 Data Preparation

This chapter covers the data preparation phase of the CRISP-DM methodology.

3.1 Select Data

Select the relevant data for the analysis.

Sampling

Apply sampling techniques to create representative subsets.

Feature Selection

Select the most relevant features for the analysis.

3.2 Clean Data

Clean the data by handling missing values, outliers, and errors.

3.3 Construct Data

Construct new features or transform existing ones.

3.4 Integrate Data

Integrate data from multiple sources.

3.5 Format Data

Format the data according to the requirements of the modeling tools.

4 Modeling

This chapter covers the modeling phase of the CRISP-DM methodology.

5 Evaluation

This chapter covers the evaluation phase of the CRISP-DM methodology.

6 Deployment

This chapter covers the deployment phase of the CRISP-DM methodology.

7 🚀 Conclusion

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